Amir Beck, Chapter 1 - Exercise 1.5

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1.5. Suppose that $\mathbb{R}^m$ and $\mathbb{R}^n$ are equipped with norms $||.||_{\mathbf{b}}$ and $||.||_{\mathbf{a}}$, respectively. Show that the induced matrix norm $||.||_{\mathbf{a,b}}$ satisfies the triangle inequality. That is, for any $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{\mathbf{m}\times\mathbf{n}}$ the inequality

$$||\mathbf{A}+\mathbf{B}||_{\mathbf{a,b}} \leq ||\mathbf{A}||_{\mathbf{a,b}} + ||\mathbf{B}||_{\mathbf{a,b}}$$

holds.

Let $A, B \in \mathbb{R}^{m\times n}$. We have the following,

$$\begin{aligned}
||A+B||_{a,b} &:= \max_{||\mathbf{x}||_a \leq 1} ||(A+B)\mathbf{x}||_b \\
&= \max_{||\mathbf{x}||_a \leq 1} ||A\mathbf{x}+B\mathbf{x}||_b \\
&\leq \max_{||\mathbf{x}||_a \leq 1} \{||A\mathbf{x}||_b + ||B\mathbf{x}||_b\} \\
&\leq \max_{||\mathbf{x}||_a \leq 1} ||A\mathbf{x}||_b + \max_{||\mathbf{x}||_a \leq 1} ||B\mathbf{x}||_b \\
&= ||A||_{a,b} + ||B||_{a,b}
\end{aligned}$$